The documentation and process conversion measures necessary to comply with this revision shall be completed by 5 JAN 93

INCH-POUND

MIL-S-19500/295B
5 October 1992
SUPERSEDING
MIL-S-19500/295A
21 May 1970

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD-EFFECT TRANSISTOR, P-CHANNEL SILICON,
TYPES 2N2608
JAN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for P-Channel, junction, silicon field-effect transistors. One level of product assurance is provided for the device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

Types	P_T 1/ T_A = 25°C	V_{GSS}	T_{STG} and T_{OP}
	mW	V dc	°C
2N2608	300	30	-65 to +200

1/ Derate linearly, 1.71 mW/°C for T_A = 25°C.

1.4 Primary electrical characteristics at T_A = 25°C.

Limit	I_{DSS} V_{DS} = -5 V dc V_{GS} = 0	$V_{GS(off)}$ V_{DS} = -5 V dc I_D = -1.0 µA dc	C_{ISS} V_{DS} = 3 V dc V_{GS} = 0 f = 1 mHz	$\|Y_{FS}\|$ V_{DS} = -5 V dc V_{GS} = 0 f = 1 kHz	I_{GSS} V_{GS} = 15 V dc V_{DS} = 0
	mA dc	V dc	pF	µmho	nA dc
Minimum	-1.0	0.75		1000	
Maximum	-5.0	6.00	10	4500	7.5

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5280 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction and physical dimensions shall be as specified in MIL-S-19500 and figure 1 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein.

3.4 Marking. Devices shall be marked in accordance with MIL-S-19500. At the option of the manufacturer, the marking of the country of origin may be omitted from the body of the transistor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500 and as specified herein. Group A inspection shall be performed on each sublot.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb, of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

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4.4.2.1 Group B inspection, table IVb of MIL-S-19500.

a. Subgroup 3, conditions for steady-state operation life are as follows: $V_{CB} \geq 10$ V dc, $T_A = 150°C$; $V_{DS} = 0$; $V_{GS} = 24$ V dc.

b. Subgroup 4, condition for bond strength is condition A.

c. Subgroup 5, not applicable.

d. Subgroup 6, condition for high-temperature life (nonoperating) is $T_{STG} = 200°C$.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

a. Subgroup 2, condition for terminal strength is test condition A.

b. Subgroups 5 and 7 not applicable.

c. Subgroup 6, condition for steady-state operation life (accelerated) are as follows:

$T_A = 150°C$; $V_{DS} = 0$; $V_{GS} = 24$ V dc.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, gate to source	3401	Bias condition C; I_G = 1 µA dc; V_{DS} = 0	$V_{(BR)GSS}$	30		V dc
Gate reverse current	3411	Bias condition C; V_{GS} = 30 V dc; V_{DS} = 0	I_{GSS1}		10	nA dc
Gate reverse current	3411	Bias condition C; V_{GS} = 15 V dc; V_{DS} = 0	I_{GSS2}		7.5	nA dc
Drain current	3413	Bias condition C, V_{GS} = 0 V dc; V_{DS} = 5 V dc	I_{DDSS1}	-1.0	-5.0	mA dc
Gate to source cutoff voltage	3403	I_D = 1 µA dc; V_{DS} = 5 V dc	$V_{GS(off)}$	0.75	6.0	V dc
Subgroup 3						
High temperature operation		T_A = +125°C				
Drain current	3413	Bias condition C, V_{GS} = 0 V dc; V_{DS} = 5 V dc	I_{DDSS2}	-0.56		mA dc
Gate reverse current	3411	Bias condition C; V_{GS} = 15 V dc; V_{DS} = 0	I_{GSS3}		15	µA dc
Low-temperature operation		T_A = -55°C				
Magnitude of small-signal, common-source short-circuit forward transfer admittance	3455	V_{DS} = 5 V dc; V_{GS} = 0 V dc; f = 1 kHz	$\|y_{fs1}\|$		4,500	µmho

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Magnitude of small-signal, common-source short-circuit forward transfer admittance	3455	V_{DS} = 5 V dc; V_{GS} = 0 V dc; f = 1 kHz	$\|y_{fs2}\|$	1,000	4,500	µmho
Small-signal, common-source short-circuit input capacitance	3431	V_{DS} = 5 V dc; V_{GS} = 0 V dc; f = 1 MHz	C_{iss}		10	pF
Common-source spot noise figure		V_{DS} = 5 V dc; V_{GS} = 0 V dc; B_W = 16%; R_G = 1 megohms; e_{gen} = 1.82 mV; R_L = 470 ohms; f = 1 kHz; (see figure 2)	NF		3	dB
Subgroups 5, 6, and						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups A, B, and C electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Gate reverse current	3411	Bias condition C; V_{GS} = 15 V dc; V_{DS} = 0	I_{GSS1}		8	nA dc
2	Gate reverse current	3411	Bias condition C; V_{GS} = 15 V dc; V_{DS} = 0	I_{GSS2}		15	nA dc
3	Drain current	3413	Bias condition C; V_{GS} = 5 V dc V_{DS} = 0	I_{DSS1}	-1.0	-5.0	mA dc
4	Drain current	3413	Bias condition C; V_{GS} = 5 V dc V_{DS} = 0	I_{DSS2}	-0.90	-5.5	mA dc
5	Magnitude of small-signal, common-source short-circuit forward transfer admittance	3455	V_{DS} = 5 V dc; V_{GS} = 0 V dc; f = 1 kHz	$\|y_{fs1}\|$	1,000	4,500	µmho
6	Magnitude of small-signal, common-source short-circuit forward transfer admittance	3455	V_{DS} = 5 V dc; V_{GS} = 0 V dc; f = 1 kHz	$\|y_{fs3}\|$	900	4,950	µmho

1/ The electrical measurements for table IVb (JANTX) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, 3, and 5.

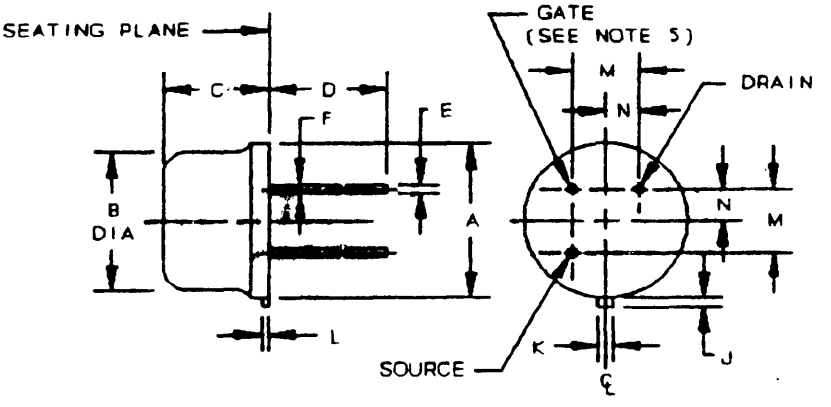
b. Subgroups 3 and 6, see table II herein, steps 2, 4, and 6.

2/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroups 2 and 3, see table II herein, steps 1, 3, and 5.

b. Subgroup 6, see table II herein, steps 2, 4, and 6.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.209	.230	5.31	5.84	
B	.178	.195	4.52	4.95	
C	.170	.210	4.32	5.33	
D	.500	.750	12.70	19.05	7
E	.016	.021	0.41	0.53	2,7
F	.016	.019	0.41	0.48	3,7
J	.028	.048	0.71	1.22	6
K	.036	.046	0.91	1.17	
L	----	.020	----	0.51	
M	.0707 Nom		1.80 Nom		4
N	.0354 Nom		0.90 Nom		4

NOTES:
1. Metric equivalents are given for general information only.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. When measured in a gauging plane .054 +.001 , - .000 (1.37 +0.03, -0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 shows the preferred measurement method.
5. The gate shall be electrically connected to the case.
6. Measured from the maximum diameter of the actual device.
7. All three leads. (see 3.3.1)
8. Diameter of leads in this zone is not controlled.

FIGURE 1. Physical dimensions.

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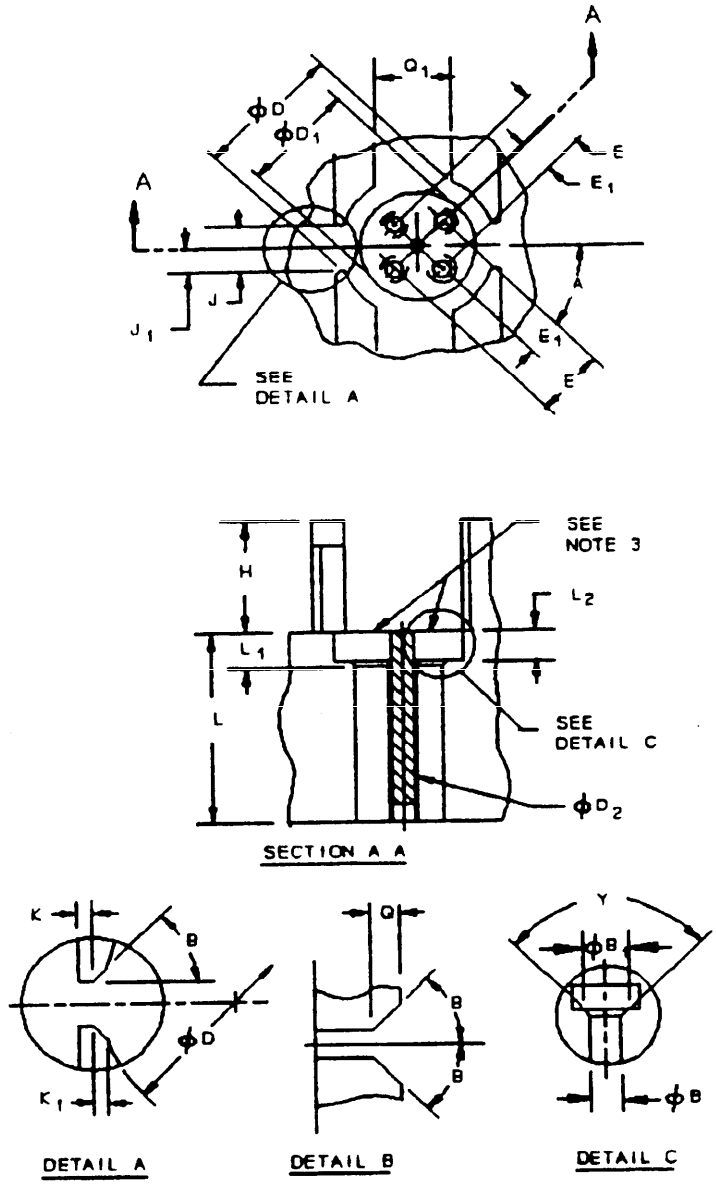


FIGURE 2. Gauge for lead and tab location for transistor type 2N2608.

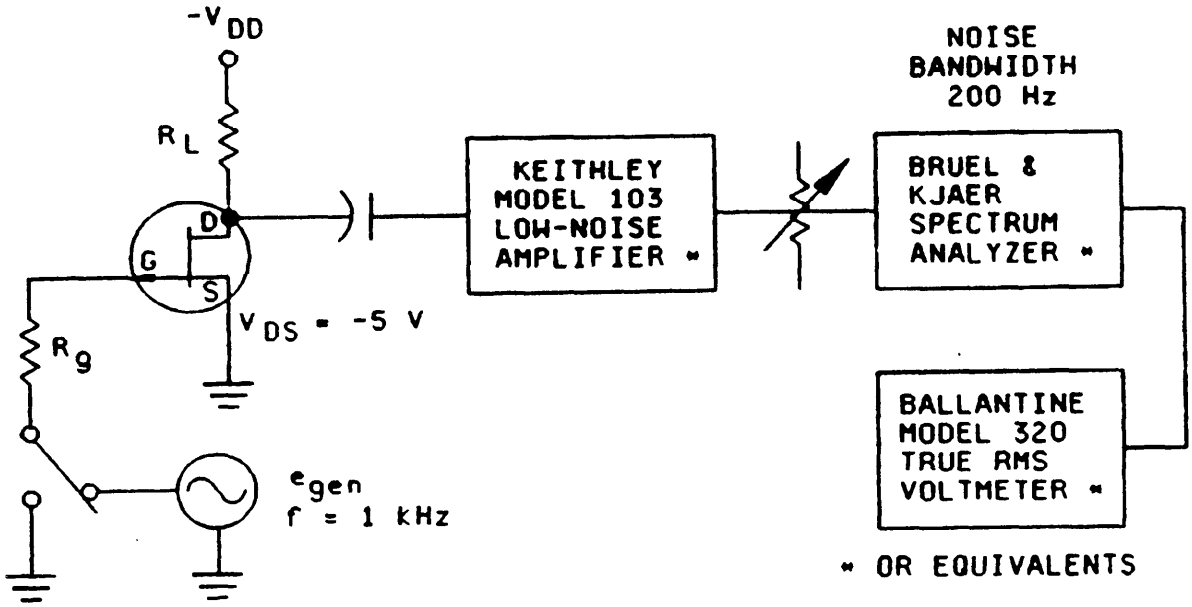
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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ϕB	.0325	.0335	0.83	0.85	4
ϕB_1	.043 Nom		1.09 Nom		4
ϕD	.2310	.2315	5.87	5.88	
ϕD_1	.159	.161	4.04	4.09	
ϕD_2	.040 Nom		1.02 Nom		5
E	.0995	.1005	2.53	2.55	
E_1	.0495	.0505	1.26	1.28	
H	.145	.155	3.68	3.94	
J	.0470	.0475	1.19	1.21	
J_1	.0235	.0245	0.60	0.62	
K	.009	.011	0.23	0.28	
K_1	.005 Nom		.127 Nom		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
L_2	.043 Nom		1.09 Nom		
Q	.040 Nom		1.02 Nom		
Q_1	.123	.127	3.12	3.23	
A	44.90°	45.10°			
B	45° Nom				
Y	90° Nom				

NOTES:
1. Metric equivalents are given for general information only.
2. The following gauge procedure shall be used: The device being measured shall be inserted until its seating plane is .125 (3.18 mm) +.010 (0.25 mm) from seating surface of the gauge. A force of 8 ±.5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not to be removed) the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible. A spacer may be used to obtain the .125 (3.18 mm) distance from the gauge seat prior to force application.
3. These surfaces to be parallel and in same plane within ±.001 (0.03 mm).
4. Four holes.
5. Pressed in.

FIGURE 2. Gauge for lead and tab location for transistor type 2N2608 - Continued.

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Procedure: The voltage and current shall be applied to the terminals, and the noise figure shall then be measured as follows:

1. Connect e_{gen} to the input of the device under test.
2. Adjust gain of system to give 0 dB reading on a convenient scale of the Ballantine voltmeter or equivalent.
3. Switch the device input to ground.
4. Increase system gain by 60 dB.
5. Read noise figure directly in dB's on Ballantine scale or equivalent.

FIGURE 3. Common-source spot noise figure test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Product assurance level and type designation.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 19
DLA - ES

User activities:
Navy - AS, CG, MC
Air Force - 13, 15

Preparing activity:
Navy - EC

Agent:
DLA - ES

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